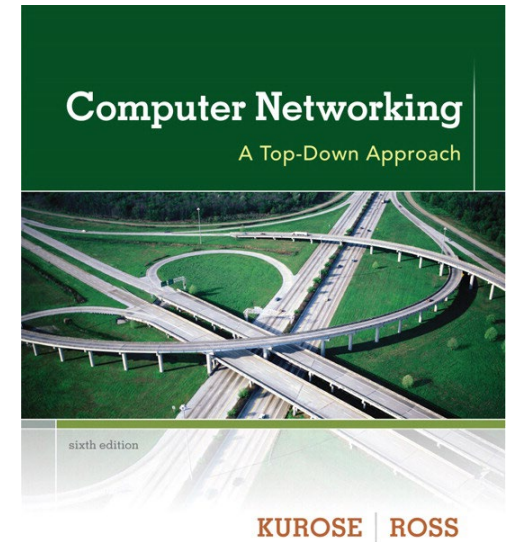


Lecture 3 (Chapter 2)

Application Layer



Computer Networking: A Top Down Approach

6th edition

Jim Kurose, Keith Ross

Addison-Wesley

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Processes communicating

process: program running within a host

- ❖ within same host, two processes communicate using **inter-process communication** (defined by OS)
- ❖ processes in different hosts communicate by exchanging **messages**

clients, servers

client process: process that initiates communication

server process: process that waits to be contacted

- ❖ aside: applications with P2P architectures have client processes & server processes

App-layer protocol defines

- ❖ types of messages exchanged,
 - e.g., request, response
- ❖ message syntax:
 - what fields in messages & how fields are described
- ❖ message semantics
 - meaning of information in fields
- ❖ rules for when and how processes send & respond to messages

open protocols:

- ❖ defined in RFCs
- ❖ allows for interoperability
- ❖ e.g., HTTP, SMTP

proprietary protocols:

- ❖ e.g., Skype

What transport service does an app need?

data integrity

- ❖ some apps (e.g., file transfer, web transactions) require 100% reliable data transfer
- ❖ other apps (e.g., audio) can tolerate some loss

timing

- ❖ some apps (e.g., Internet telephony, interactive games) require low delay to be “effective”

throughput

- ❖ some apps (e.g., multimedia) require minimum amount of throughput to be “effective”
- ❖ other apps (“elastic apps”) make use of whatever throughput they get

security

- ❖ encryption, data integrity,
...

Transport service requirements: common apps

application	data loss	throughput	time sensitive
file transfer	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5kbps-1Mbps video: 10kbps-5Mbps	yes, 100' s msec
stored audio/video	loss-tolerant	same as above	
interactive games	loss-tolerant	few kbps up	yes, few secs
text messaging	no loss	elastic	yes, 100' s msec yes and no

Internet apps: application, transport protocols

application	application layer protocol	underlying transport protocol
e-mail	SMTP [RFC 2821]	TCP
remote terminal access	Telnet [RFC 854]	TCP
Web	HTTP [RFC 2616]	TCP
file transfer	FTP [RFC 959]	TCP
streaming multimedia	HTTP (e.g., YouTube), RTP [RFC 1889]	TCP or UDP
Internet telephony	RTP, proprietary (e.g., Skype)	TCP or UDP

Chapter 2: outline

2.1 Introduction

2.2 Web and HTTP

2.3 FTP

2.4 electronic mail

- SMTP, POP3, IMAP

2.5 DNS

2.6 P2P applications

2.7 socket programming
with UDP and TCP

Web and HTTP

First, a review...

- ❖ *web page* consists of *objects*
- ❖ object can be HTML file, JPEG image, Java applet, audio file,...
- ❖ web page consists of *base HTML-file* which includes *several referenced objects*
- ❖ each object is addressable by a *URL*, e.g.,

`www.someschool.edu/someDept/pic.gif`

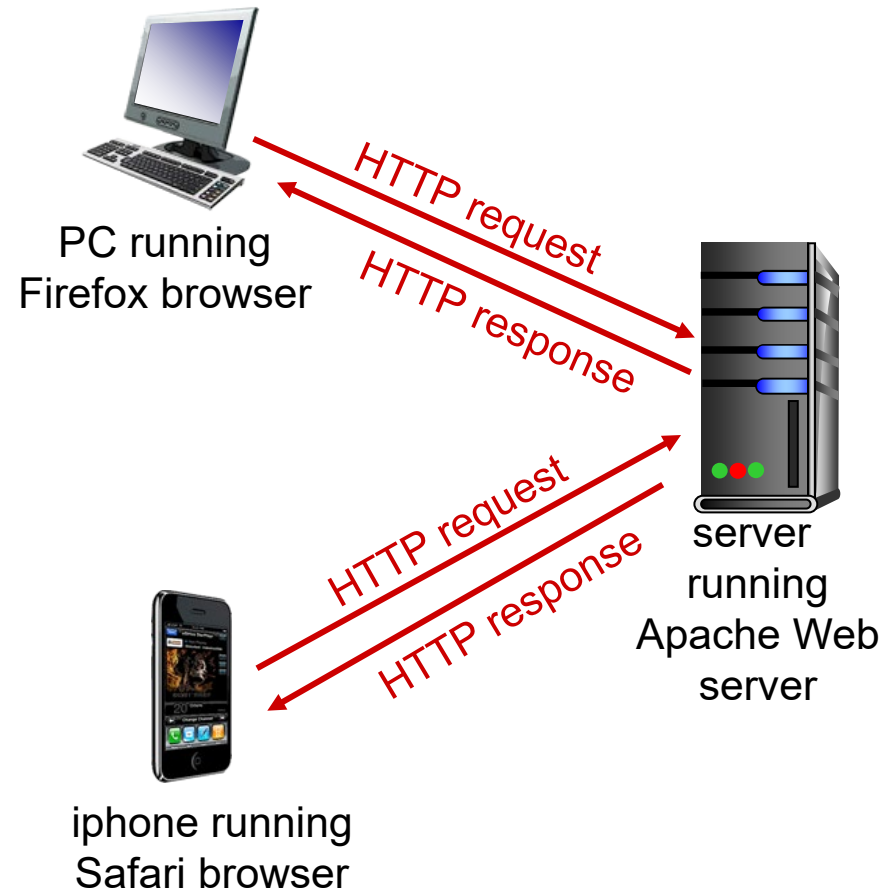
host name

path name

HTTP overview

HTTP: hypertext transfer protocol

- ❖ Web's application layer protocol
- ❖ client/server model
 - **client**: browser that requests, receives, (using HTTP protocol) and "displays" Web objects
 - **server**: Web server sends (using HTTP protocol) objects in response to requests



HTTP overview (continued)

uses TCP:

- ❖ client initiates TCP connection (creates socket) to server, port 80
- ❖ server accepts TCP connection from client
- ❖ HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- ❖ TCP connection closed

HTTP is “stateless”

- ❖ server maintains no information about past client requests

HTTP connections

non-persistent HTTP

- ❖ at most one object sent over TCP connection
 - connection then closed
- ❖ downloading multiple objects required multiple connections

persistent HTTP

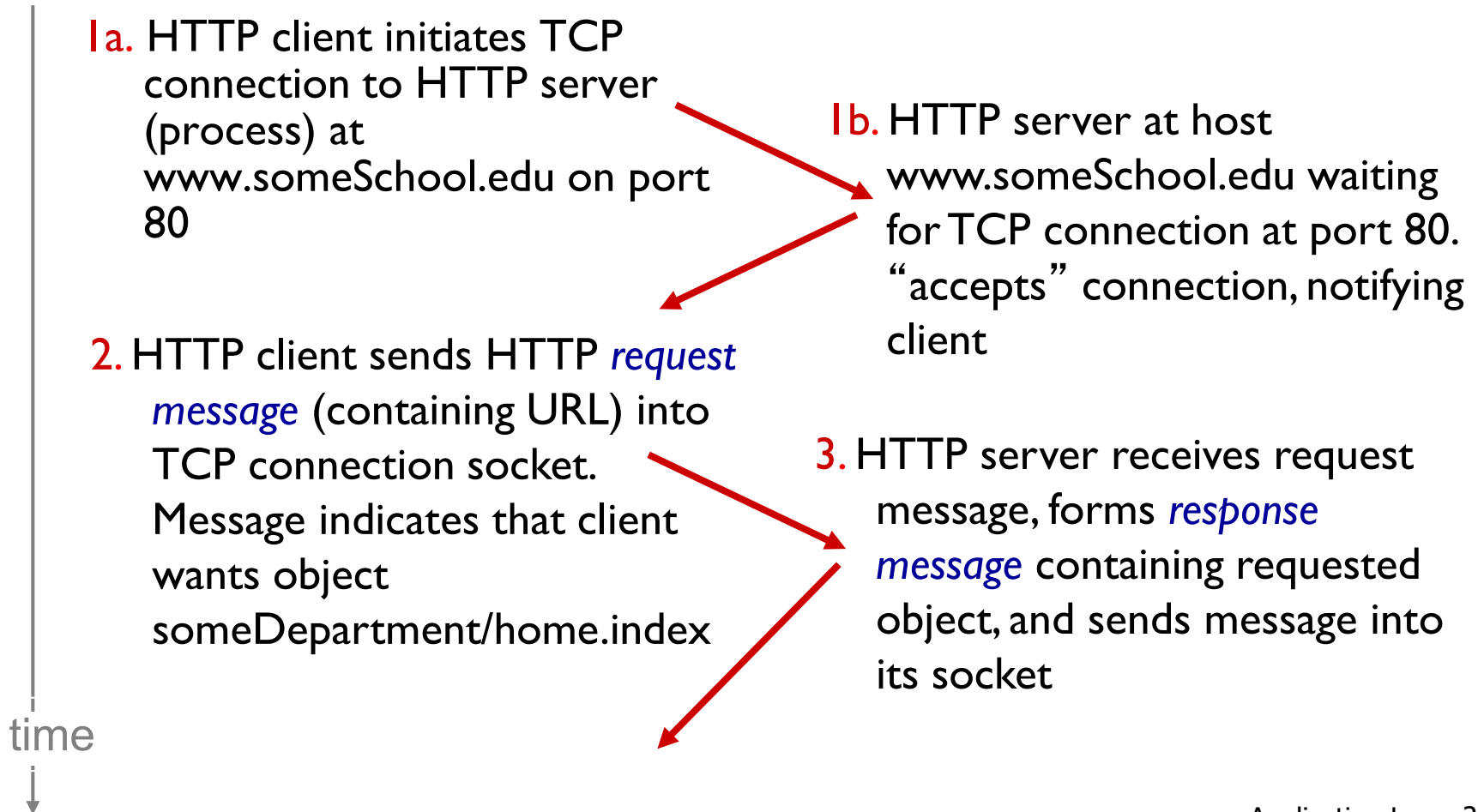
- ❖ multiple objects can be sent over single TCP connection between client, server

Non-persistent HTTP

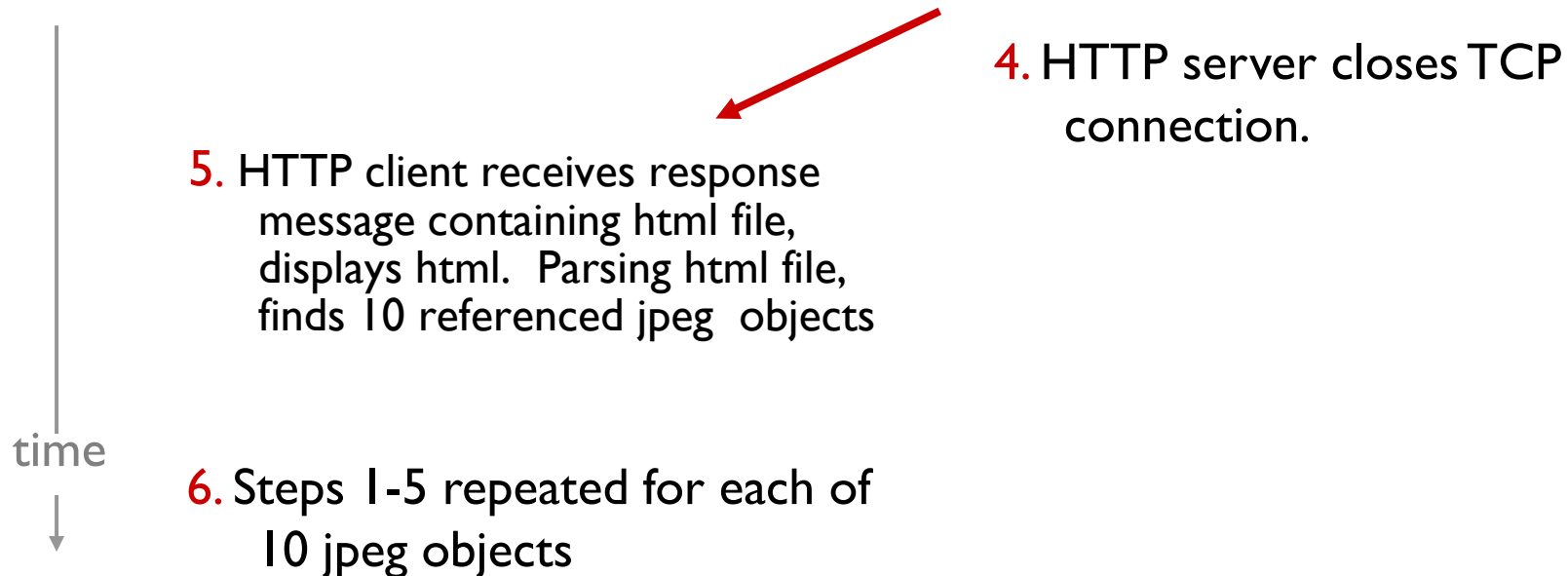
suppose user enters URL:

`www.someSchool.edu/someDepartment/home.index`

(contains text,
references to 10
jpeg images)



Non-persistent HTTP (cont.)

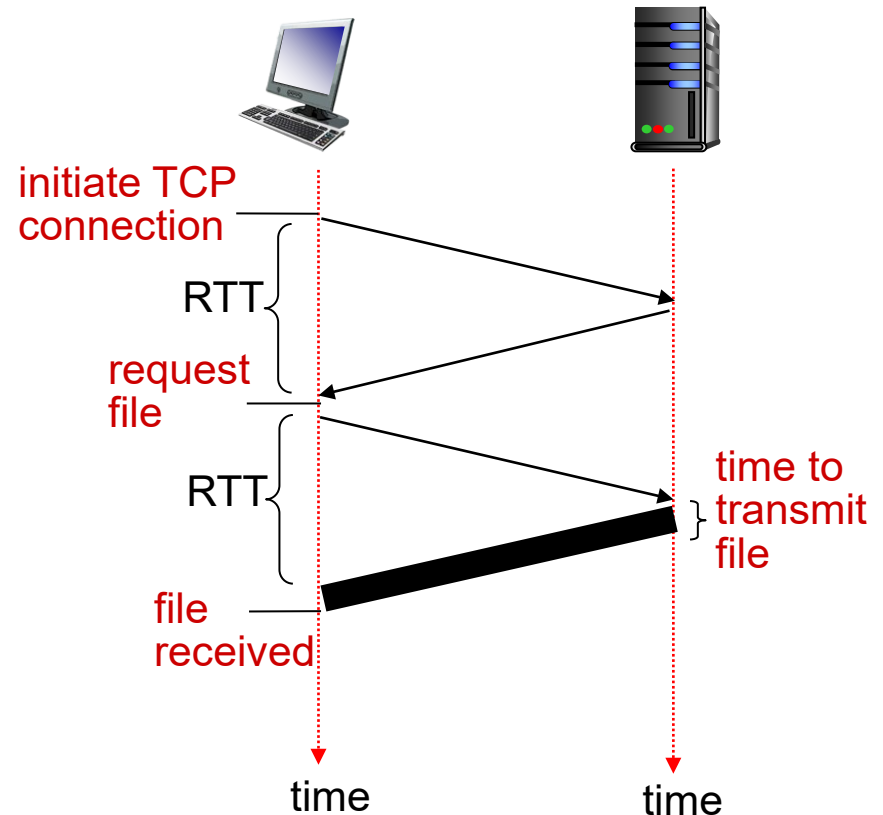


Non-persistent HTTP: response time

RTT (definition): time for a small packet to travel from client to server and back

HTTP response time:

- ❖ one RTT to initiate TCP connection
- ❖ one RTT for HTTP request and first few bytes of HTTP response to return
- ❖ file transmission time
- ❖ non-persistent HTTP response time =
 $2\text{RTT} + \text{file transmission time}$



Persistent HTTP

non-persistent HTTP issues:

- ❖ requires 2 RTTs per object
- ❖ overhead for *each* TCP connection
- ❖ browsers often open parallel TCP connections to fetch referenced objects

persistent HTTP:

- ❖ server leaves connection open after sending response
- ❖ subsequent HTTP messages between same client/server sent over open connection
- ❖ client sends requests as soon as it encounters a referenced object
- ❖ as little as one RTT for all the referenced objects

HTTP request message

- ❖ two types of HTTP messages: *request, response*
- ❖ **HTTP request message:**
 - ASCII (human-readable format)

request line
(GET, POST,
HEAD commands)

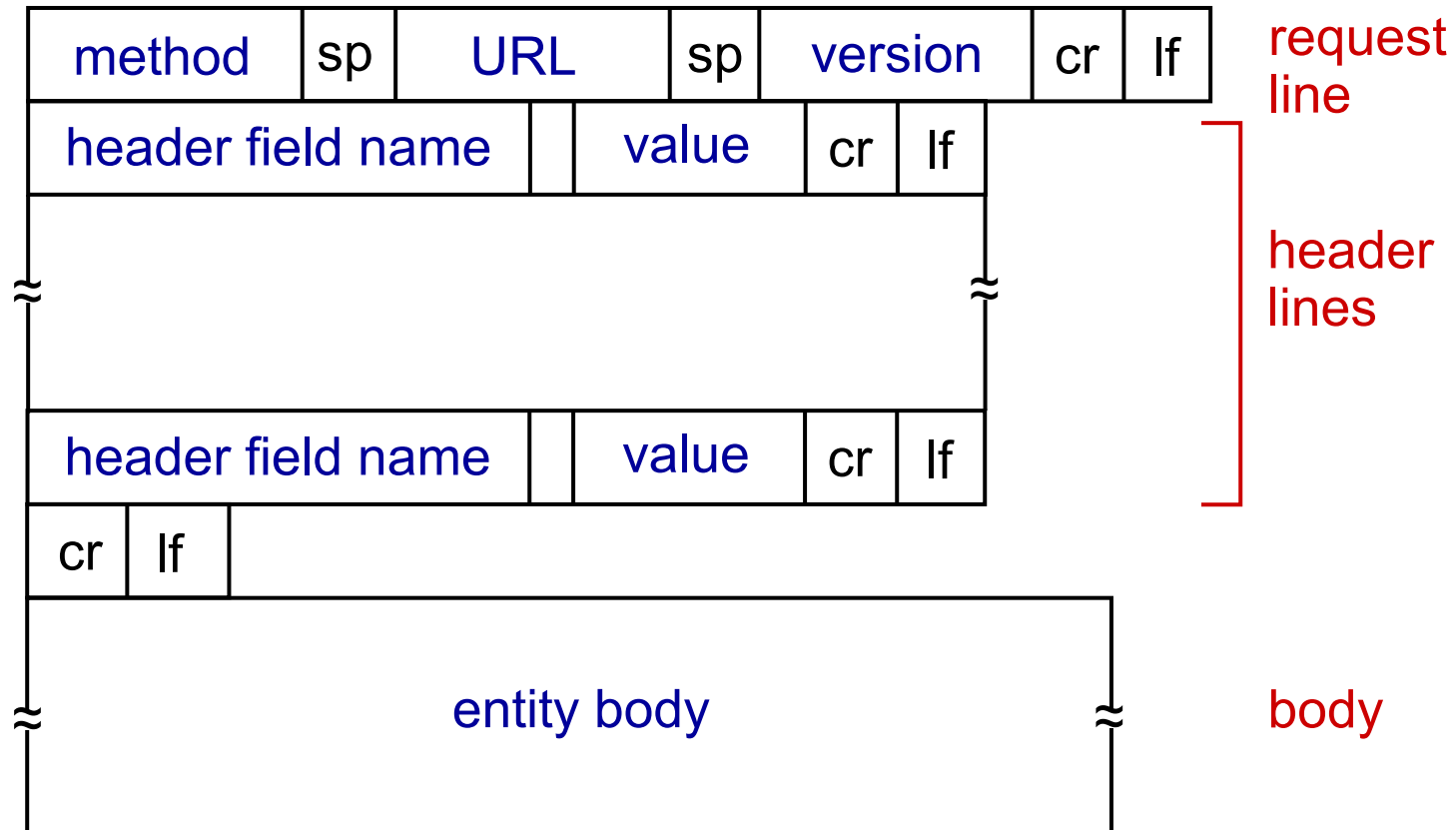
header
lines

carriage return,
line feed at start
of line indicates
end of header lines

```
GET /index.html HTTP/1.1\r\n
Host: www-net.cs.umass.edu\r\n
User-Agent: Firefox/3.6.10\r\n
Accept: text/html,application/xhtml+xml\r\n
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip,deflate\r\n
Accept-Charset: ISO-8859-1,utf-8;q=0.7\r\n
Keep-Alive: 115\r\n
Connection: keep-alive\r\n
\r\n
```

carriage return character
line-feed character

HTTP request message: general format



Uploading form input

POST method:

- ❖ web page often includes form input
- ❖ input is uploaded to server in entity body

URL method:

- ❖ uses GET method
- ❖ input is uploaded in URL field of request line:

`www.somesite.com/animalsearch?monkeys&banana`

HTTP response message

status line
(protocol
status code
status phrase)

header
lines

data, e.g.,
requested
HTML file

```
HTTP/1.1 200 OK\r\n
Date: Sun, 26 Sep 2010 20:09:20 GMT\r\n
Server: Apache/2.0.52 (CentOS)\r\n
Last-Modified: Tue, 30 Oct 2007 17:00:02
      GMT\r\n
ETag: "17dc6-a5c-bf716880"\r\n
Accept-Ranges: bytes\r\n
Content-Length: 2652\r\n
Keep-Alive: timeout=10, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=ISO-8859-
      1\r\n
\r\n
data data data data data ...
```

HTTP response status codes

- ❖ status code appears in 1st line in server-to-client response message.
- ❖ some sample codes:

200 OK

- request succeeded, requested object later in this msg

301 Moved Permanently

- requested object moved, new location specified later in this msg (Location:)

400 Bad Request

- request msg not understood by server

404 Not Found

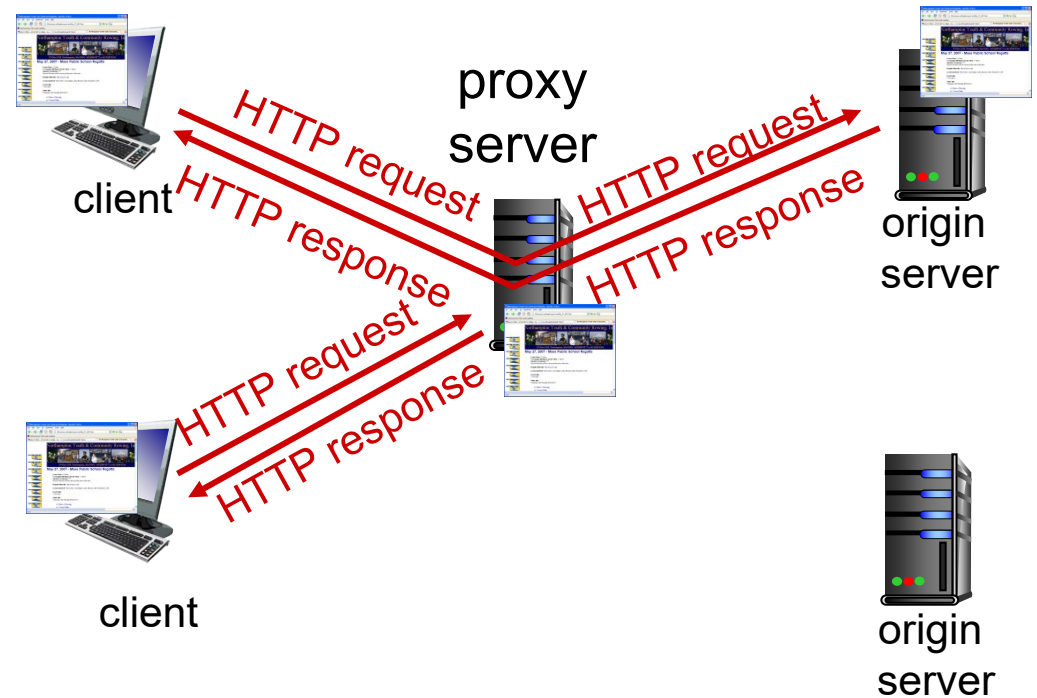
- requested document not found on this server

505 HTTP Version Not Supported

Web caches (proxy server)

goal: satisfy client request without involving origin server

- ❖ user sets browser: Web accesses via cache
- ❖ browser sends all HTTP requests to cache
 - object in cache: cache returns object
 - else cache requests object from origin server, then returns object to client



More about Web caching

- ❖ cache acts as both client and server
 - server for original requesting client
 - client to origin server
- ❖ typically cache is installed by ISP (university, company, residential ISP)

why Web caching?

- ❖ reduce response time for client request
- ❖ reduce traffic on an institution's access link
- ❖ Internet dense with caches: enables “poor” content providers to effectively deliver content (so too does P2P file sharing)

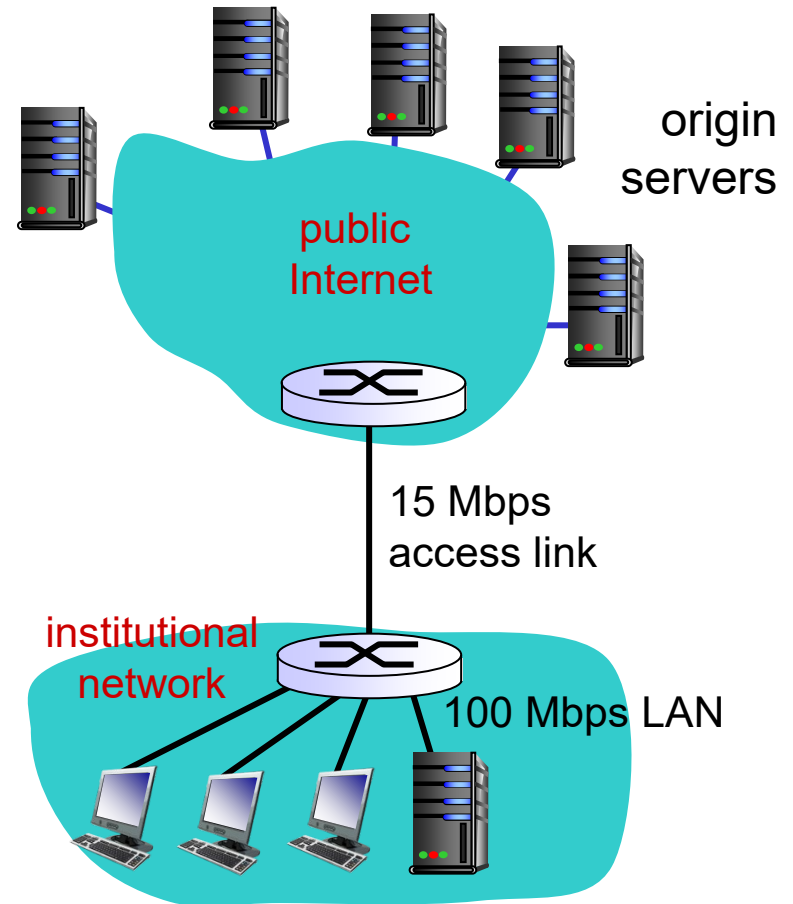
Caching example:

assumptions:

- ❖ avg object size: 1M bits
- ❖ avg request rate from browsers to origin servers: 15/sec
- ❖ RTT from internet side router to any origin server: 2 sec
- ❖ access link rate: 15 Mbps

consequences:

- ❖ LAN utilization: 15%
- ❖ access link utilization = **100%** *problem!*
- ❖ total delay = Internet delay + access delay + LAN delay
= 2 sec + minutes + usecs



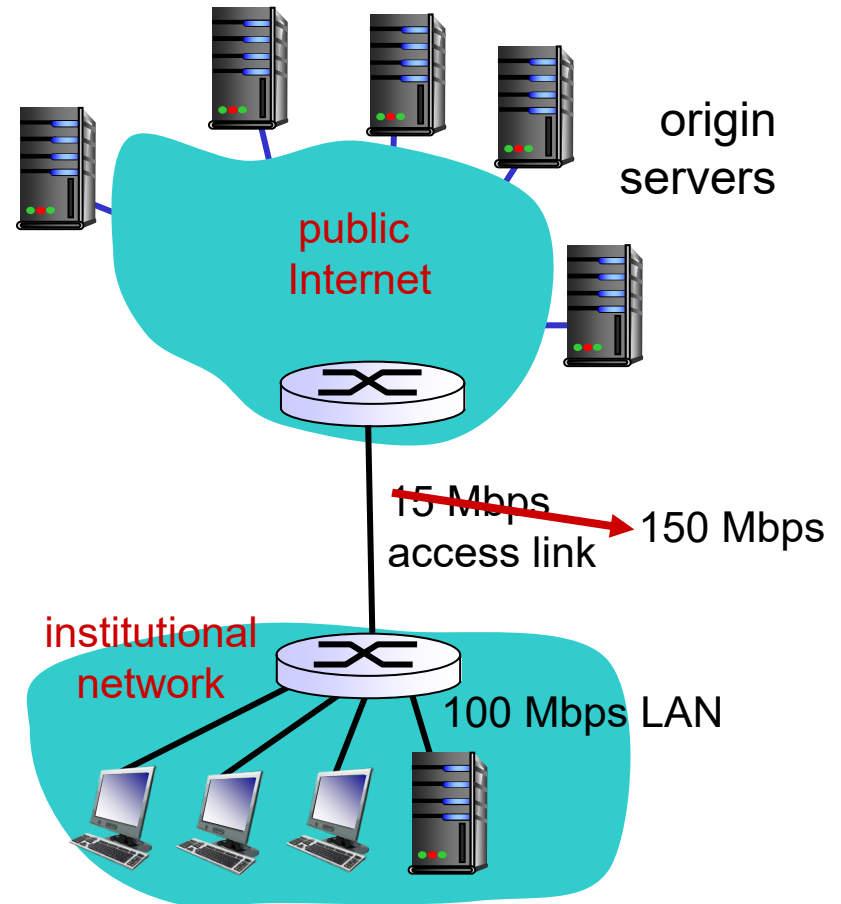
Caching example: fatter access link

assumptions:

- ❖ avg object size: 1M bits
- ❖ avg request rate from browsers to origin servers: 15/sec
- ❖ RTT from institutional router to any origin server: 2 sec
- ❖ access link rate: ~~15 Mbps~~ → 150 Mbps

consequences:

- ❖ LAN utilization: 15%
- ❖ access link utilization = ~~100%~~ 10%
- ❖ total delay = Internet delay + access delay + LAN delay
= 2 sec + ~~minutes~~ usecs
msecs



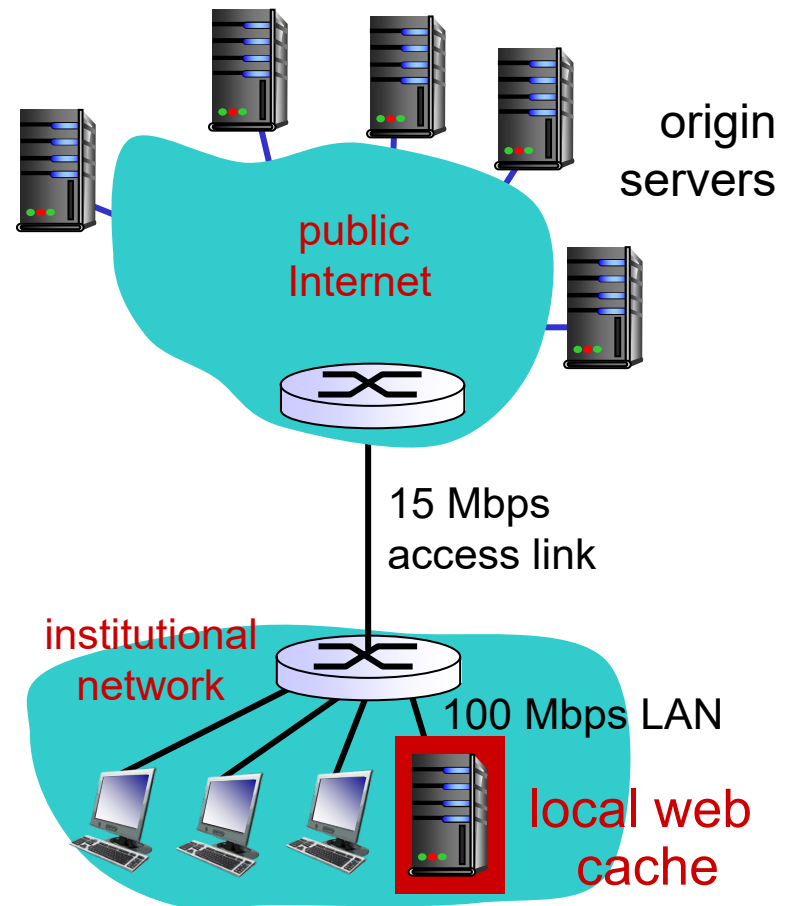
Cost: increased access link speed (not cheap!)

Caching example: install local cache

assumptions:

- ❖ avg object size: 1M bits
- ❖ avg request rate from browsers to origin servers: 15/sec
- ❖ avg data rate to browsers
- ❖ RTT from institutional router to any origin server: 2 sec
- ❖ access link rate: 15 Mbps

Cost: web cache (cheap!)



Chapter 2: outline

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2.3 FTP

2.4 electronic mail

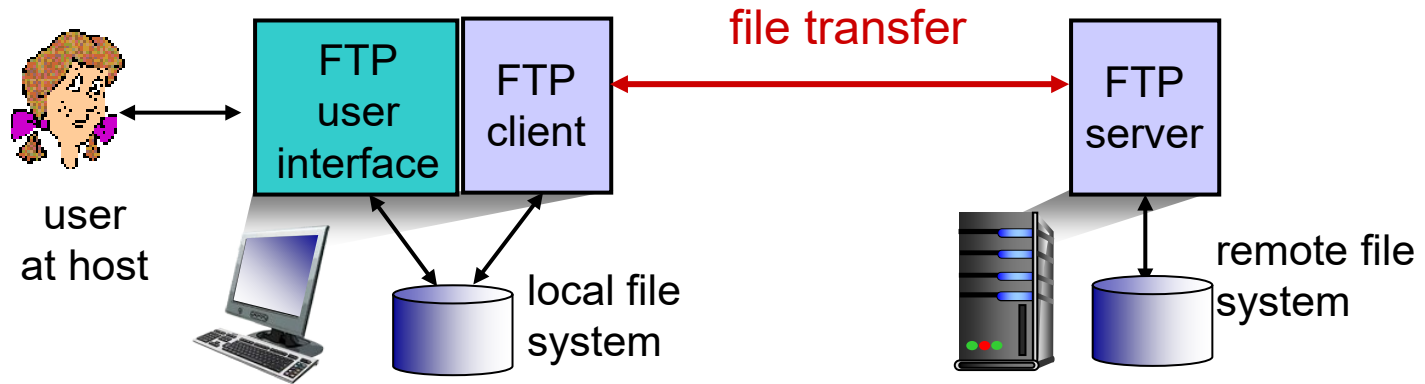
- SMTP, POP3, IMAP

2.5 DNS

2.6 P2P applications

2.7 socket programming
with UDP and TCP

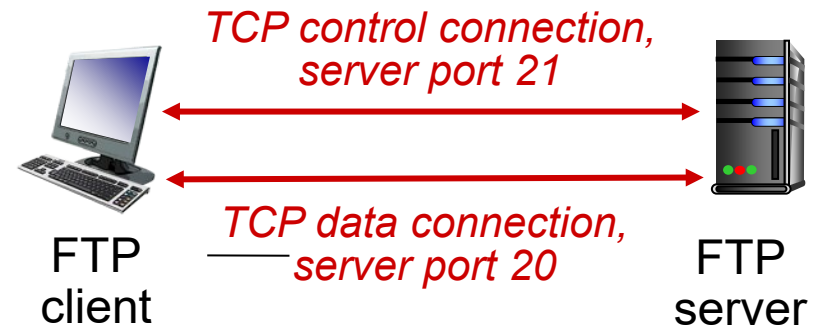
FTP: the file transfer protocol



- ❖ transfer file to/from remote host
- ❖ client/server model
 - *client*: side that initiates transfer (either to/from remote)
 - *server*: remote host
- ❖ ftp: RFC 959
- ❖ ftp server: port 21

FTP: separate control, data connections

- ❖ FTP client contacts FTP server at port 21, using TCP
- ❖ client authorized over control connection
- ❖ client browses remote directory, sends commands over control connection
- ❖ when server receives file transfer command, **server** opens 2nd TCP data connection (for file) to client
- ❖ after transferring one file, server closes data connection



- ❖ server opens another TCP data connection to transfer another file
- ❖ control connection: **“out of band”**
- ❖ FTP server maintains “state”: current directory, earlier authentication

FTP commands, responses

sample commands:

- ❖ sent as ASCII text over control channel
- ❖ **USER *username***
- ❖ **PASS *password***
- ❖ **LIST** return list of file in current directory
- ❖ **RETR *filename*** retrieves (gets) file
- ❖ **STOR *filename*** stores (puts) file onto remote host

sample return codes

- ❖ status code and phrase (as in HTTP)
- ❖ **331 Username OK, password required**
- ❖ **125 data connection already open; transfer starting**
- ❖ **425 Can't open data connection**
- ❖ **452 Error writing file**

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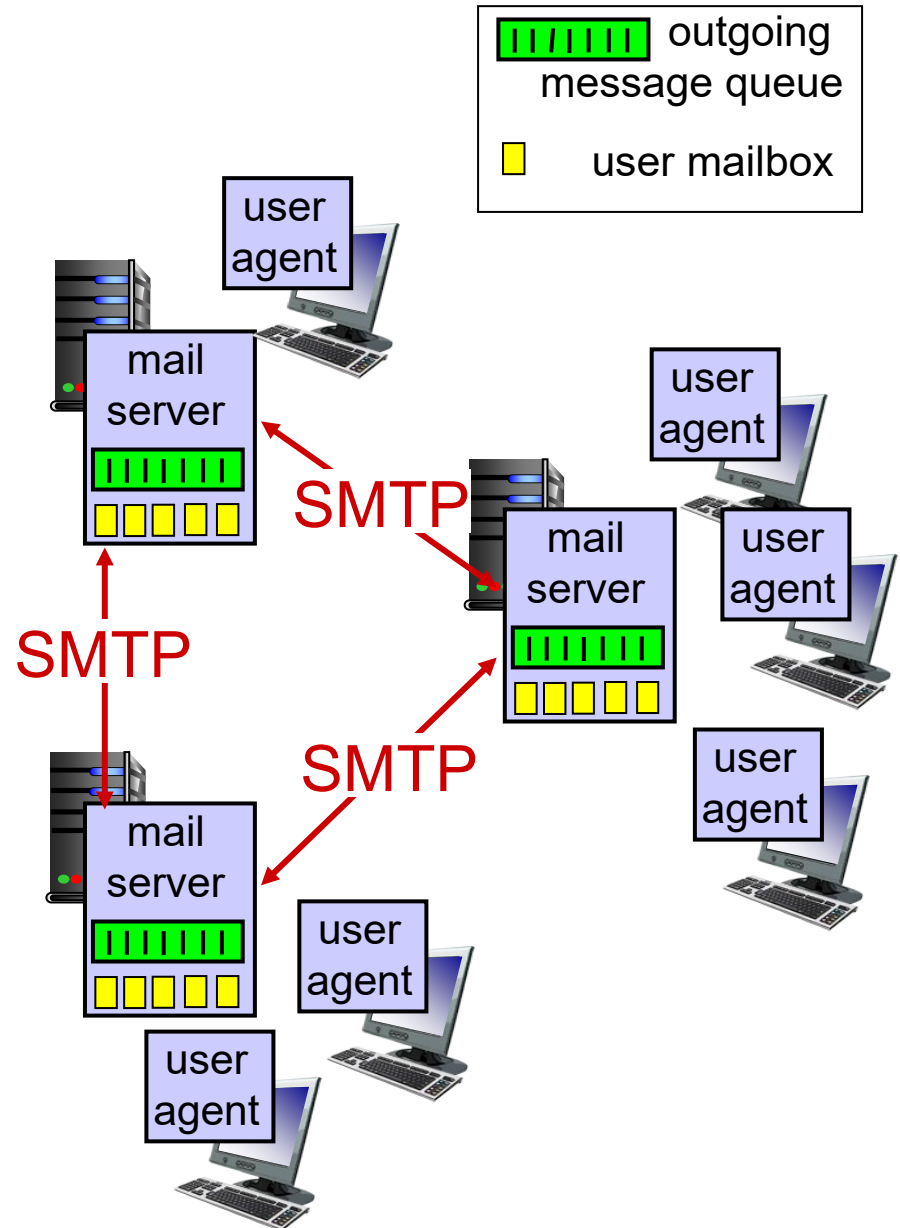
Electronic mail

Three major components:

- ❖ user agents
- ❖ mail servers
- ❖ simple mail transfer protocol: SMTP

User Agent

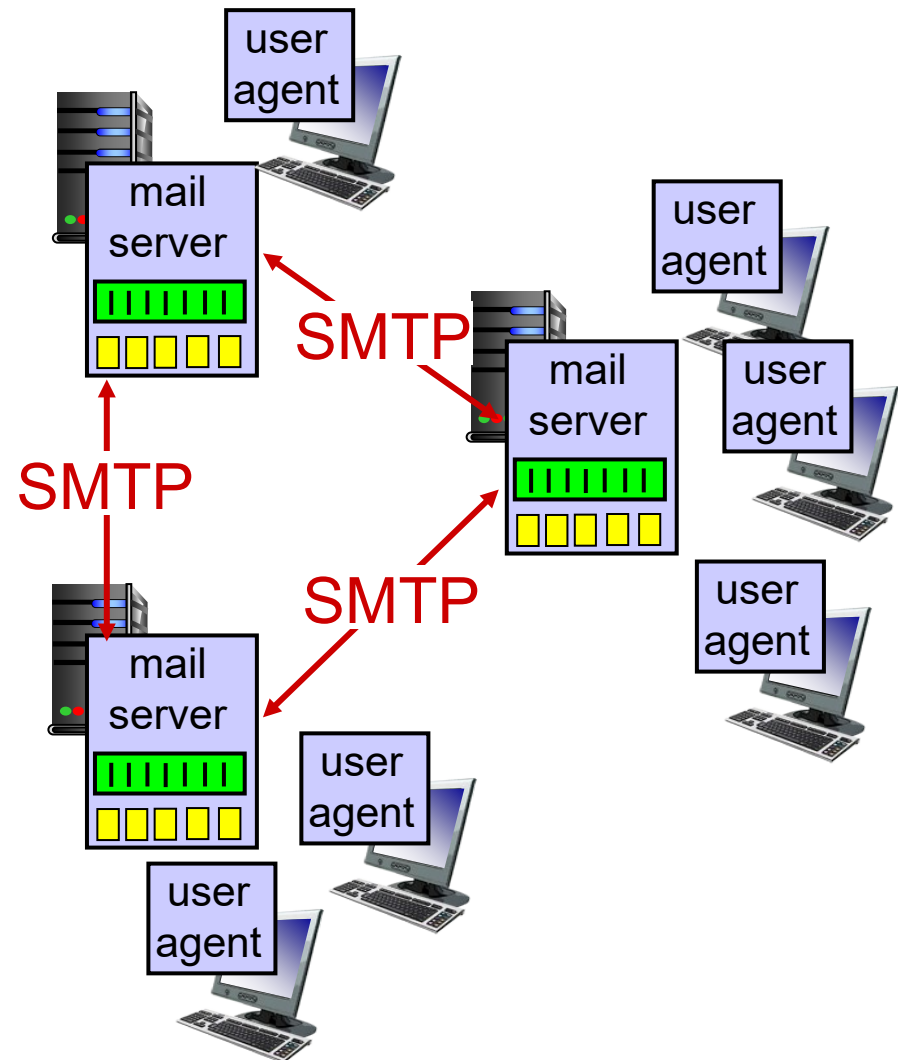
- ❖ a.k.a. “mail reader”
- ❖ composing, editing, reading mail messages
- ❖ e.g., Outlook, Thunderbird, iPhone mail client
- ❖ outgoing, incoming messages stored on server



Electronic mail: mail servers

mail servers:

- ❖ *mailbox* contains incoming messages for user
- ❖ *message queue* of outgoing (to be sent) mail messages
- ❖ *SMTP protocol* between mail servers to send email messages
 - client: sending mail server
 - “server”: receiving mail server

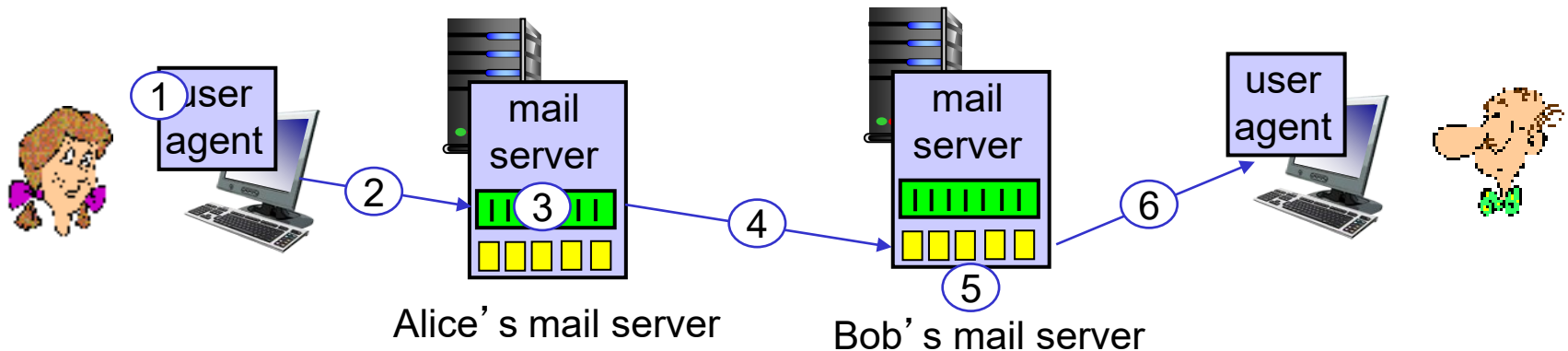


Electronic Mail: SMTP [RFC 2821]

- ❖ uses TCP to reliably transfer email message from client to server, port 25
- ❖ direct transfer: sending server to receiving server
- ❖ three phases of transfer
 - handshaking (greeting)
 - transfer of messages
 - closure
- ❖ command/response interaction (like HTTP, FTP)
 - **commands:** ASCII text
 - **response:** status code and phrase
- ❖ messages must be in 7-bit ASCII

Scenario: Alice sends message to Bob

- 1) Alice uses UA to compose message “to”
`bob@someschool.edu`
- 2) Alice’s UA sends message to her mail server; message placed in message queue
- 3) client side of SMTP opens TCP connection with Bob’s mail server
- 4) SMTP client sends Alice’s message over the TCP connection
- 5) Bob’s mail server places the message in Bob’s mailbox
- 6) Bob invokes his user agent to read message



Sample SMTP interaction

```
S: 220 hamburger.edu
C: HELO crepes.fr
S: 250 Hello crepes.fr, pleased to meet you
C: MAIL FROM: <alice@crepes.fr>
S: 250 alice@crepes.fr... Sender ok
C: RCPT TO: <bob@hamburger.edu>
S: 250 bob@hamburger.edu ... Recipient ok
C: DATA
S: 354 Enter mail, end with "." on a line by itself
C: Do you like ketchup?
C: How about pickles?
C: .
S: 250 Message accepted for delivery
C: QUIT
S: 221 hamburger.edu closing connection
```

Mail message format

SMTP: protocol for exchanging email msgs

RFC 822: standard for text message format:

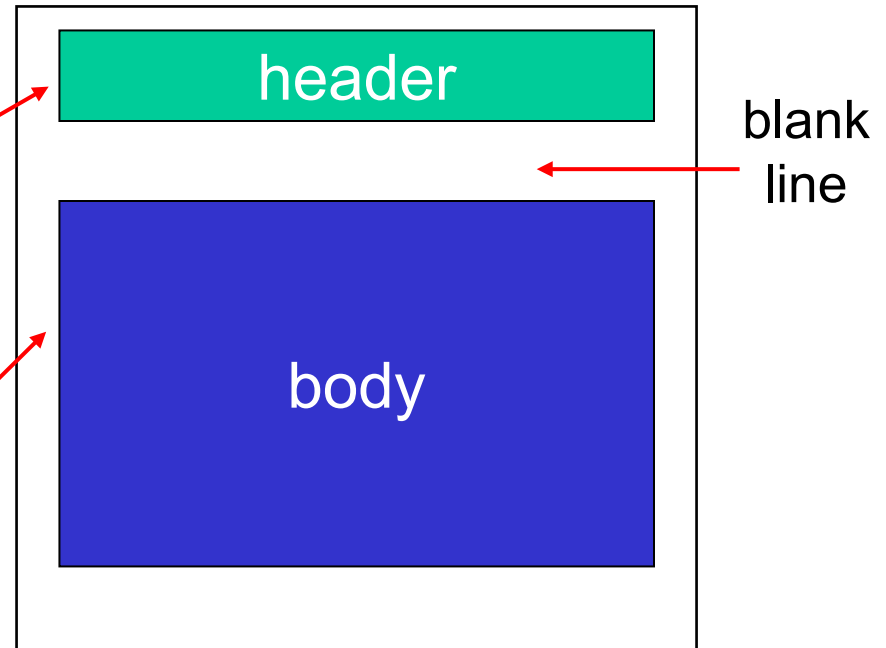
❖ header lines, e.g.,

- To:
- From:
- Subject:

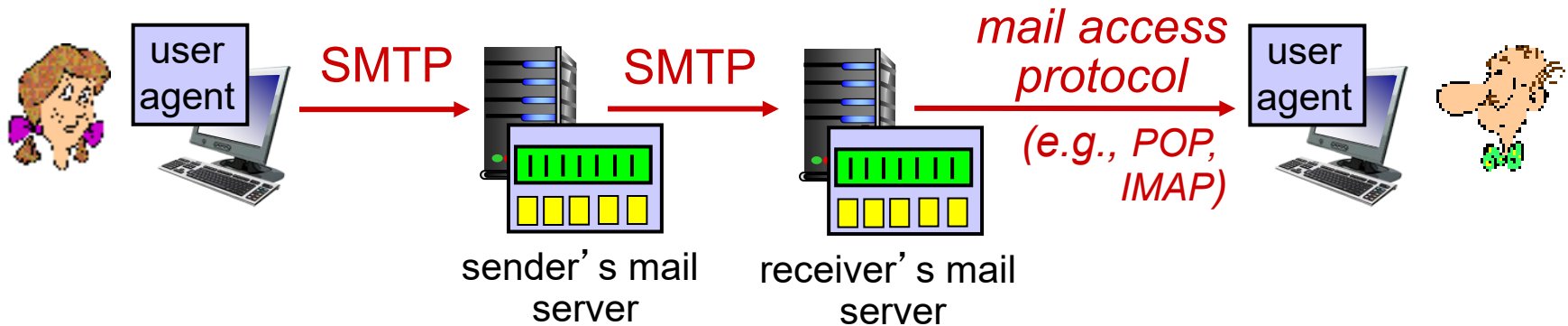
different from SMTP MAIL
FROM, RCPT TO:
commands!

❖ Body: the “message”

- ASCII characters only



Mail access protocols



- ❖ **SMTP**: delivery/storage to receiver's server
- ❖ mail access protocol: retrieval from server
 - **POP**: Post Office Protocol [RFC 1939]: authorization, download
 - **IMAP**: Internet Mail Access Protocol [RFC 1730]: more features, including manipulation of stored msgs on server
 - **HTTP**: gmail, Hotmail, Yahoo! Mail, etc.

POP3 protocol

authorization phase

- ❖ client commands:
 - **user**: declare username
 - **pass**: password
- ❖ server responses
 - **+OK**
 - **-ERR**

```
S: +OK POP3 server ready
C: user bob
S: +OK
C: pass hungry
S: +OK user successfully logged on
```

transaction phase, client:

- ❖ **list**: list message numbers
- ❖ **retr**: retrieve message by number
- ❖ **dele**: delete
- ❖ **quit**

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: <message 1 contents>
S: .
C: dele 1
C: retr 2
S: <message 2 contents>
S: .
C: dele 2
C: quit
S: +OK POP3 server signing off
```

POP3 (more) and IMAP

more about POP3

- ❖ previous example uses POP3 “download and delete” mode
 - Bob cannot re-read e-mail if he changes client
- ❖ POP3 “download-and-keep”: copies of messages on different clients
- ❖ POP3 is stateless across sessions

IMAP

- ❖ keeps all messages in one place: at server
- ❖ allows user to organize messages in folders
- ❖ keeps user state across sessions:
 - names of folders and mappings between message IDs and folder name

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DNS: domain name system

people: many identifiers:

- name, passport #

Internet hosts, routers:

- IP address (32 bit) - used for addressing datagrams
- “name”, e.g., `www.yahoo.com` - used by humans

Q: how to map between IP address and name, and vice versa ?

Domain Name System:

- ❖ *distributed database*
implemented in hierarchy of many *name servers*
- ❖ *application-layer protocol*: hosts, name servers communicate to *resolve* names (address/name translation)
 - note: core Internet function, implemented as application-layer protocol
 - complexity at network's “edge”

DNS: services, structure

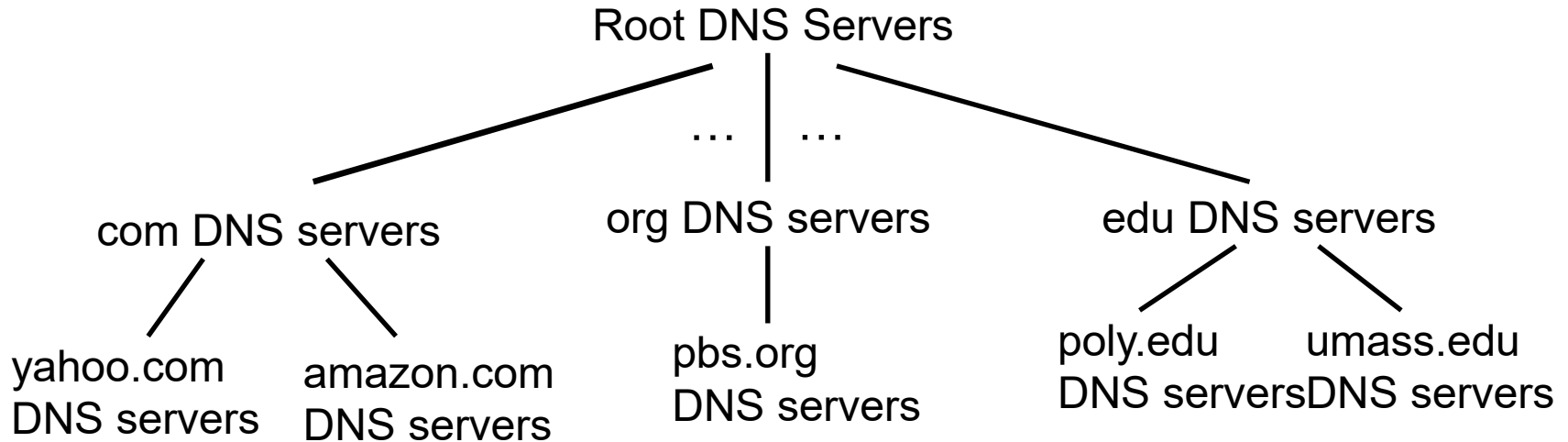
DNS services

- ❖ hostname to IP address translation
- ❖ host aliasing
 - alias names
- ❖ mail server aliasing
- ❖ load distribution
 - replicated Web servers: many IP addresses correspond to one name

why not centralize DNS?

- ❖ single point of failure
- ❖ traffic volume
- ❖ distant centralized database
- ❖ maintenance

DNS: a distributed, hierarchical database

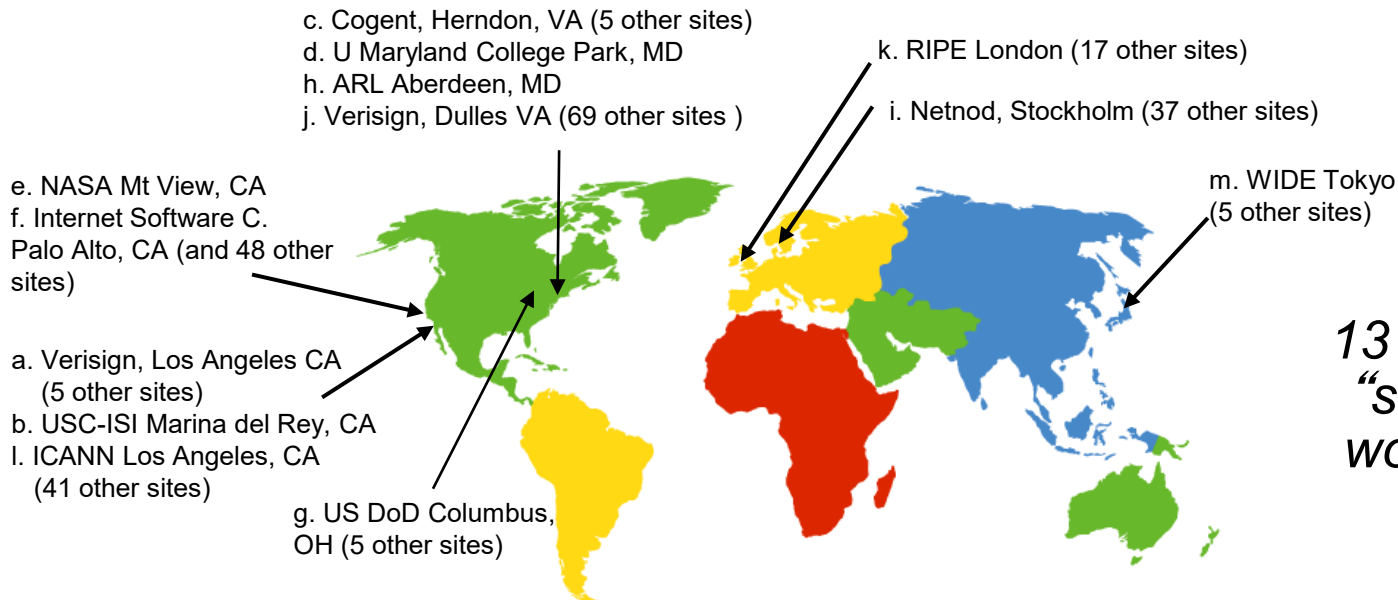


client wants IP for www.amazon.com; 1st approx:

- ❖ client queries root server to find com DNS server
- ❖ client queries .com DNS server to get amazon.com DNS server
- ❖ client queries amazon.com DNS server to get IP address for www.amazon.com

DNS: root name servers

- ❖ contacted by local name server that can not resolve name
- ❖ root name server:
 - contacts authoritative name server if name mapping not known
 - gets mapping
 - returns mapping to local name server



*13 root name
“servers”
worldwide*

TLD, authoritative servers

top-level domain (TLD) servers:

- responsible for com, org, net, edu, aero, jobs, museums, and all top-level country domains, e.g.: uk, fr, ca, jp
- Network Solutions maintains servers for .com TLD
- Educause for .edu TLD

authoritative DNS servers:

- organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts
- can be maintained by organization or service provider

Local DNS name server

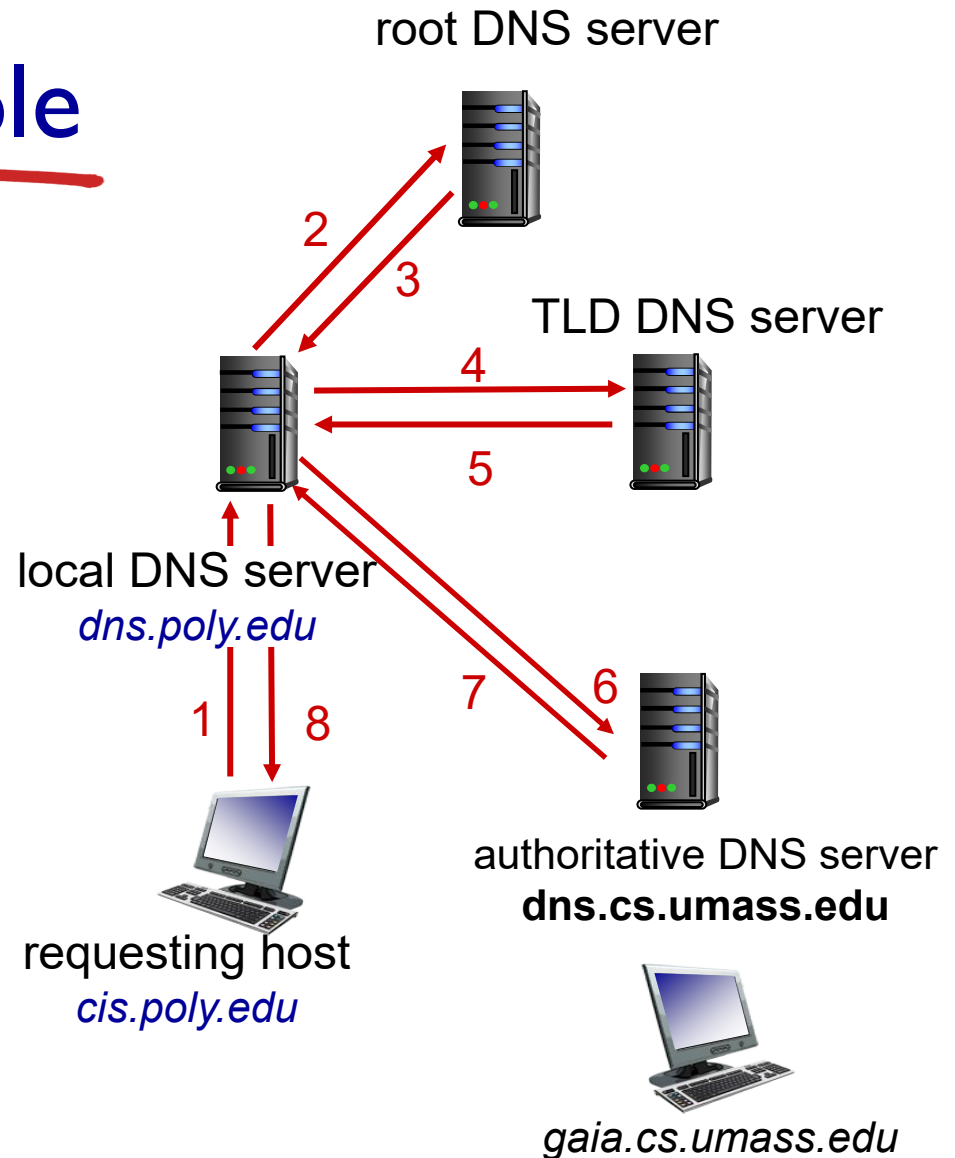
- ❖ does not strictly belong to hierarchy
- ❖ each ISP (residential ISP, company, university) has one
 - also called “default name server”
- ❖ when host makes DNS query, query is sent to its local DNS server
 - has local cache of recent name-to-address translation pairs (but may be out of date!)
 - acts as proxy, forwards query into hierarchy

DNS name resolution example

- ❖ host at cis.poly.edu wants IP address for gaia.cs.umass.edu

iterated query:

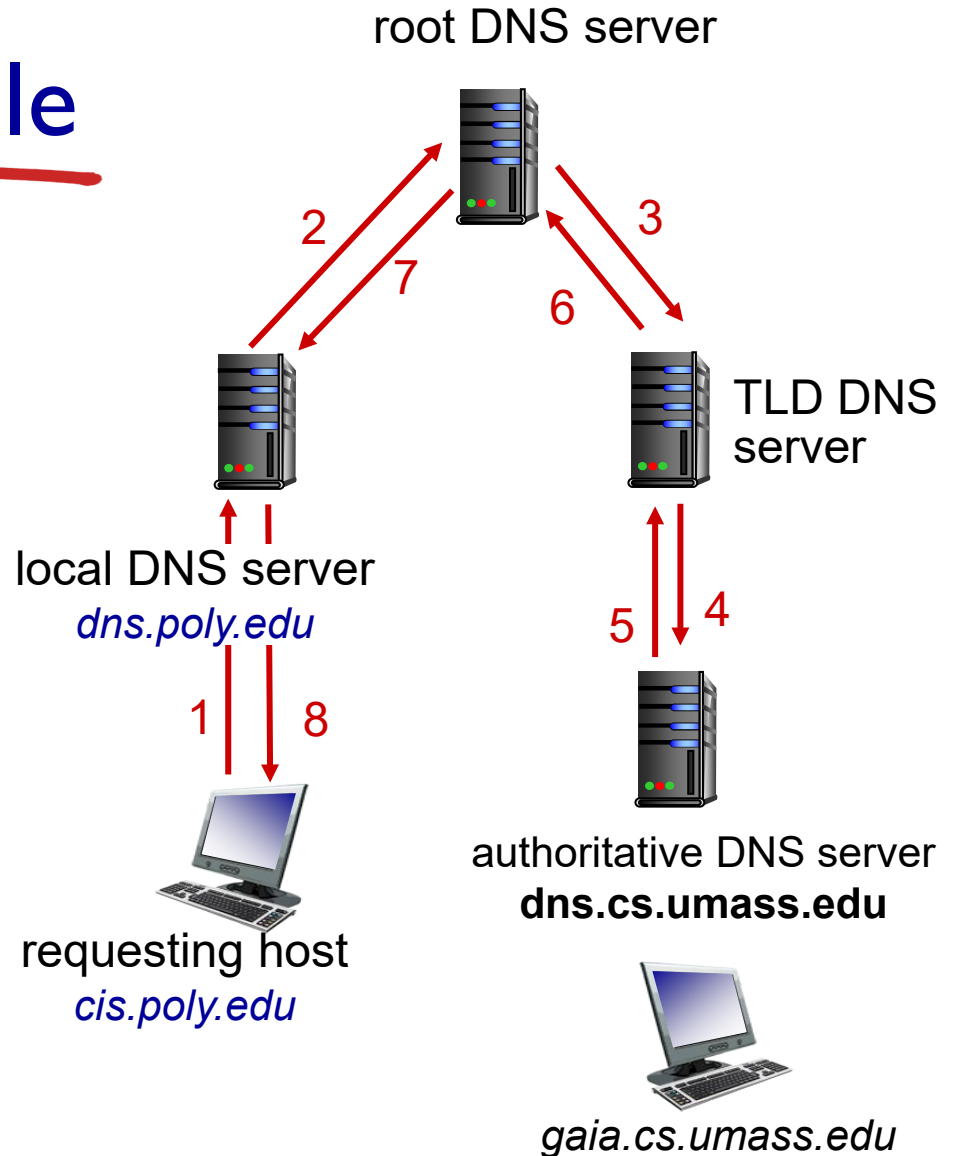
- ❖ contacted server replies with name of server to contact
- ❖ “I don’t know this name, but ask this server”



DNS name resolution example

recursive query:

- ❖ puts burden of name resolution on contacted name server
- ❖ heavy load at upper levels of hierarchy?



DNS: caching, updating records

- ❖ once (any) name server learns mapping, it *caches* mapping
 - cache entries timeout (disappear) after some time (TTL)
 - TLD servers typically cached in local name servers
 - thus root name servers not often visited
- ❖ cached entries may be *out-of-date* (best effort name-to-address translation!)
 - if name host changes IP address, may not be known Internet-wide until all TTLs expire
- ❖ update/notify mechanisms proposed IETF standard
 - RFC 2136

DNS records

DNS: distributed db storing resource records (RR)

RR format: (name, value, type, ttl)

type=A

- **name** is hostname
- **value** is IP address

type=NS

- **name** is domain (e.g., foo.com)
- **value** is hostname of authoritative name server for this domain

type=CNAME

- **name** is alias name for some “canonical” (the real) name
- **www.ibm.com** is really **servereast.backup2.ibm.com**
- **value** is canonical name

type=MX

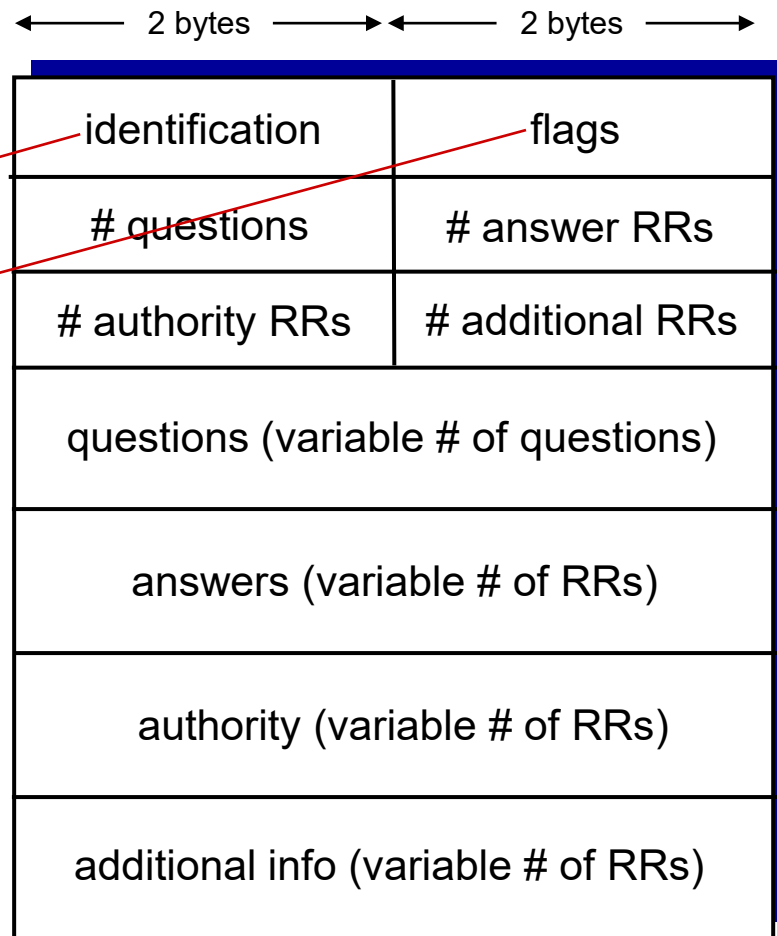
- **value** is name of mailserver associated with **name**

DNS protocol, messages

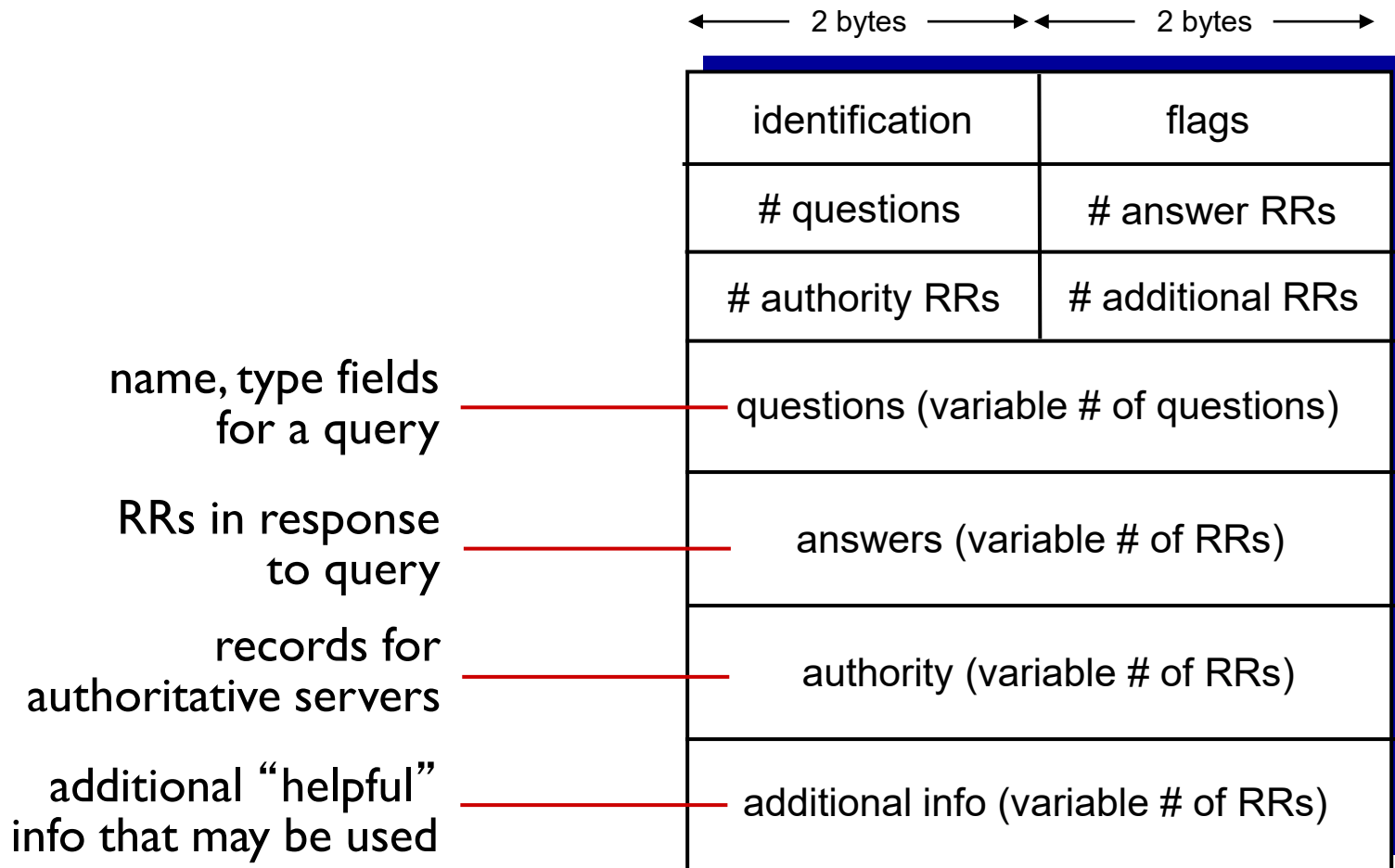
- ❖ *query* and *reply* messages, both with same *message format*

msg header

- ❖ **identification:** 16 bit # for query, reply to query uses same #
- ❖ **flags:**
 - query or reply
 - recursion desired
 - recursion available
 - reply is authoritative



DNS protocol, messages



Inserting records into DNS

- ❖ example: new startup “Network Utopia”
- ❖ register name networkutopia.com at *DNS registrar* (e.g., Network Solutions)
 - provide names, IP addresses of authoritative name server (primary and secondary)
 - registrar inserts two RRs into .com TLD server:
(networkutopia.com, dns1.networkutopia.com, NS)
(dns1.networkutopia.com, 212.212.212.1, A)

Attacking DNS

DDoS attacks

- ❖ Bombard root servers with traffic
 - Not successful to date
 - Traffic Filtering
 - Local DNS servers cache IPs of TLD servers, allowing root server bypass
- ❖ Bombard TLD servers
 - Potentially more dangerous

Redirect attacks

- ❖ Man-in-middle
 - Intercept queries
- ❖ DNS poisoning
 - Send bogus replies to DNS server, which caches